🧠 What is Azure Firewall?

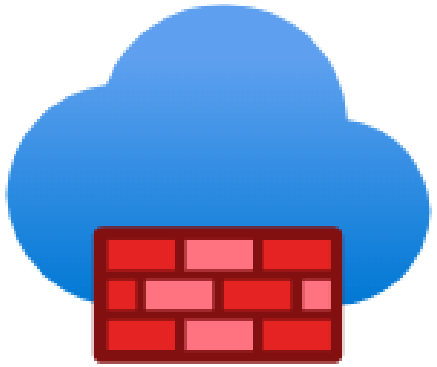
Azure Firewall is a **managed Layer 3–7 security service** in Azure.

Unlike NSG:

- NSG → Layer 4 (Port based)
- Azure Firewall → Layer 7 (Application + FQDN filtering)

It provides:

✔ Centralized traffic control
✔ Outbound internet filtering
✔ Inbound DNAT
✔ Application rules (FQDN-based)
✔ Threat intelligence filtering
✔ Logging & monitoring



Firewalls

🏗️ Architecture Requirement

Azure Firewall must be deployed in:

```yaml
Subnet name: AzureFirewallSubnet
```

Minimum size:

```
/26
```

Example:

```yaml
VNet: 10.0.0.0/16

AzureFirewallSubnet : 10.0.0.0/26
WorkloadSubnet      : 10.0.1.0/24
```

🔥 Types of Azure Firewall

Type	Use Case
Standard	Most common
Premium	TLS inspection, IDPS
Basic	Small lab / low budget

For AZ-104 → Standard is enough.

🔥 Azure Firewall Rule Types

1️⃣ Network Rules (Layer 4)

Based on:

- IP
- Port
- Protocol (TCP/UDP)

Example:

Allow 8.8.8.8 port 53

2️⃣ Application Rules (Layer 7)

Based on:

- FQDN
- HTTP/HTTPS

Example:

Allow access to:

```
www.microsoft.com
```

3️⃣ NAT Rules

Used for:

- DNAT
- Inbound internet → private VM

🔥 LAB – Deploy Azure Firewall

STEP 1 – Create VNet

Create VNet with:

Subnet	Address
AzureFirewallSubnet	10.0.0.0/26
WorkloadSubnet	10.0.1.0/24

Important:

Subnet name must be exactly:

```nginx
AzureFirewallSubnet
```

STEP 2 – Deploy Azure Firewall

Portal → Search → Azure Firewall → Create

Fill:

- Name: AZ-FW
- Region: Same as VNet
- Firewall tier: Standard
- Virtual Network: Select VNet
- Public IP: Create new

Deploy.

STEP 3 – Create Route Table

Create Route Table:

Add route:

Address Prefix	**0.0.0.0/0**
Next Hop Type	Virtual Appliance
Next Hop IP	Firewall Private IP

Attach route table to:

👉 WorkloadSubnet

STEP 4 – Create Rule Collection

Go to:

Firewall → Rules → Add Rule Collection

◆ Application Rule Example

Allow only Microsoft website:

Rule Type: Application

Source: 10.0.1.0/24

Protocol: HTTP, HTTPS

Target FQDN:

```
www.microsoft.com
```

Action: Allow

Network Rule Example

Allow DNS:

Destination:

```
8.8.8.8
```

Port:

```
53
```

Protocol: UDP

STEP 5 – Test

From Workload VM:

```powershell
curl www.microsoft.com    ✅ Works
curl www.google.com       ❌ Blocked
```

🔥 DNAT Lab (Inbound)

Goal:

Internet → Firewall Public IP → Internal VM (RDP)

Create NAT Rule:

Setting	Value
Destination	Firewall Public IP
Destination Port	3389
Translated Address	10.0.1.4
Translated Port	3389

Now RDP to:

```vbnet
Firewall Public IP
```

Traffic forwarded to internal VM.

🔥 NSG vs Azure Firewall

NSG	Azure Firewall
Layer 4	Layer 7
Free	Paid
Subnet/NIC	Centralized
No FQDN filtering	FQDN filtering